

Q → 1) 10.10.10.0/12

$$n = 12 - 8 = 4$$

Subnet mask = 255.255.0.0

$$\text{total network} = 2^n = 2^4 = 16$$

network

$$\text{total IP per network} = 2^n = 2^4 = 16$$

network

~~total IP per network = 2^n = 2^4 = 16~~

network  
20

10.10.10.0  
10.10.10.1 → network IP  
10.10.10.2  
10.10.10.3  
...  
Host IP

~~10.10.10.15~~ → broadcast IP  
10.0.0.1048575

2) 192.168.2.0/30

$$n = 30 - 24 = 6$$

Subnet mask = 255.255.255.252

$$128 + 64 + 32 + 16 + 8 + 4 = 252$$

$$\text{total network} = 2^n = 2^6 = 64$$

$$\text{total IP per network} = 2^n = 4 \text{ IP}$$

$$\text{total hosts} = 2^7 - 2 = 2^4 - 2 = 2$$

192.168.2.0  $\rightarrow$  Network IP

192.168.2.1

192.168.2.2

}  $\rightarrow$  Host IP

192.168.2.3

$\rightarrow$  Broadcast IP

3) 190.168.2.0 / 19

$$n = 19 - 16 = 3$$

Subnet mask = 255.255.224.0

~~before~~ before  $n = 3$

$$\text{total network} = 2^n = 2^3 = 8$$

$$\text{remain } n = 32 - 19 = 13$$

$$\text{total IP per network} = 2^{13} = 8192$$

192.168.2.0

$\rightarrow$  Network IP

192.168.2.1

192.168.2.2

} Host IP

192.168.2.31

$\rightarrow$  Broadcast IP